Optics & Laser Technology

**Date:** November 2024

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# Effects of He-Ar shielding gas compositions on arc plasma physical properties in rotating laser+GMAW hybrid fillet welding: Numerical simulation

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 10.1016/j.optlastec.2024.111231



## Highlights:

Outline

- A numerical model is established for rotating laser hybrid fillet welding with He-Ar shielding gas compositions.
- Effects of rotating laser and He-Ar shielding gas compositions on arc physical properties are investigated.
- With the He content increases, the arc temperature and distribution are more concentrated, the arc flow velocity and pressure decrease.
- He-Ar shielding gas mixtures increase the arc heat flux density, suppress the shielding effect of laser-induced plume, and improve the weld penetration.

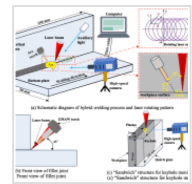
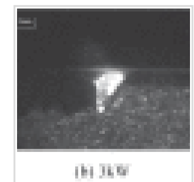
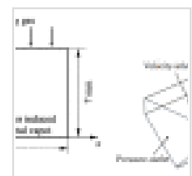
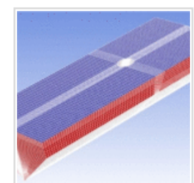
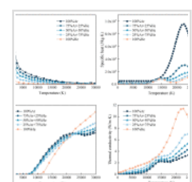
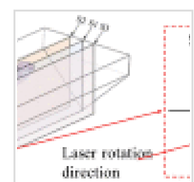


## Abstract

To understand the influence mechanism of He-Ar shielding gas composition on arc physical properties in laser+GMAW hybrid welding, this study establishes an arc numerical analysis model in rotating laser+GMAW hybrid fillet welding by considering the geometric feature of joint configuration and the physical parameters of He-Ar shielding gas. The effects of He-Ar shielding gas mixtures in different ratios on arc plasma temperature, velocity, and pressure are investigated, respectively. The results show that with the increasing He content in the shielding gas, the overall shape of arc is compressed, the peak temperature decreases, the arc temperature distribution range is more concentrated, which is beneficial to increasing the weld penetration depth. The arc plasma flow velocity increases with the He content increases, but the plasma flow pattern becomes disordered when the He content reaches 50%, which is not conducive to the stability of arc. The arc pressure value and the high-pressure distribution region near the workpiece surface increase significantly with an increasing He content ratio, due to the contraction effect of high He content mixed shielding gas on the arc.

## Keywords

Figures (15)




Tables (3)

He-Ar shielding gas mixture; Rotating laser-arc hybrid welding; Plasma physical properties; Numerical simulation; Fillet welding



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## 1. Introduction

Laser-arc hybrid welding combines the dual advantages of laser and arc welding [1], which has received widespread attention in welding aluminum alloys [2], due to the higher welding speed, deeper weld penetration, good weld formation, small post-weld deformation, and strong bridging ability [3], [4], [5]. However, there are many process parameters of laser-arc hybrid welding, and if they are not properly matched, welding defects such as undercut, porosity, and crack may occur. Compared with laser-arc hybrid welding, laser oscillation can significantly change the dynamic behavior characteristics of keyhole and molten pool [6], [7], which can obviously reduce defects such as porosity, undercut, and hump, thus receiving increased attention from scholars [8], [9]. However, laser oscillation also leads to a large decrease in weld penetration, especially for the materials with high conductivity such as aluminum alloy, which is detrimental to the weld quality. During welding process, shielding gas not only can effectively prevent harmful gases in the air from entering molten pool, but also affects the arc physical properties, weld penetration and weld width [10]. He-Ar shielding gas mixtures have a positive effect on increasing the weld penetration and improving the welding process stability, which effectively combines the advantages of both inert gases [11]. However, the arc physical properties become more complex due to the rotation of laser, the geometrical characteristics of joint and the ratio of He-Ar shielding gas mixture. Therefore, further studies about the influence mechanism of He-Ar shielding gas composition on the arc physical properties of rotating laser-MIG hybrid welding are needed.

Currently, an extensive effort has made on the welding process with mixed shielding gases. Previous studies are focused on arc or laser welding process through experiments and there is still lack of study refer to the arc physical properties in rotating laser-arc hybrid welding, especially for the complex fillet joint. Kuk et al. [12] investigated the fatigue life of 5083 aluminum alloy joints under various temperatures and shielding gas mixtures and found that the fatigue strength of the welded joints increased with increasing He gas ratio. The dilution was best and higher fatigue strength of weld metal specimens were observed when 33%Ar+67%He shielding gas mixture was used. Zhang et al. [13] examined the corrosion resistance of joints with 2%$O_2$ and 2%$N_2$ added to argon, respectively. Their results demonstrated that the corrosion resistance of joints was significantly enhanced in the 98% Ar+2% $N_2$ mixed shielding gas. However, Aar et al. [14] discovered that adding a certain amount of $H_2$ to the Ar shielding gas decreased the mechanical properties of joint. Thus, only an appropriate mixed shielding gas composition can enhance the mechanical properties of joints.

Lei et al. [15] investigated the effects of 100%Ar and He-Ar shielding gas mixture on the porosity defects of aluminum alloy laser welding. They concluded that under the He-Ar shielding gas mixture, the number of plasmas decreased, the stability of keyhole improved, the reliability of weld increased, and the porosity was reduced to less than 1%. Similarly, Cai et al. [16] pointed out that with the increase of He content, the hybrid plasma size decreased, and the plasma is more easily attracted to the keyhole, which ultimately affected the effective laser power density and the weld penetration depth. Pandya et al. [17] discovered that the weld depth-width ratio of TIG and A-TIG welding significantly increased by adding 1%$H_2$ to pure Ar, due to hydrogen's higher thermal conductivity, which improves melting efficiency. Zhu et al. [18] investigated the effects of various He contents on weld formation and metal transfer behavior in rotary arc assisted by laser and demonstrated that with an increasing He content, the weld penetration increased, and the weld formation improved initially but deteriorated thereafter. The welding formation was optimal due to the waveforms of welding current and arc voltage exhibited minimal fluctuation when the He content in the shielding gas was 30%, and the transition behavior of metal droplets is a stable rotating spray transfer. Cai et al. [19], [20] concluded that the arc voltage and sidewall fusion depth increase with the increasing He content, the arc length decreases, and arc core expands with the increasing $CO_2$ content. When the ratio of shielding gas is 80%Ar+10%$CO_2$+10%He, the welding process is stable, and the weld quality is relatively good. In addition, when used He-Ar alternating shielding gas, the droplet transfer modes are short-circuiting transition and spherical transition, which periodically affect the weld penetration depth [21]. Pavan et al. [22] pointed out that the helium-containing arc generated greater melting depth due to the concentrated shape and higher heat flux density for thick 316L plate A-TIG welding under varying shielding gas. Although the above experimental results suggest that the mixed shielding gas has an obvious positive impact on enhancing the mechanical properties of joint and weld formation, as well as suppressing weld defects, the high cost of He and trial and error experiments require a lot of time and cycles. Furthermore, these experiments do not fully elucidate the mechanism of interaction between the arc physical characteristics and the parameters of welding shielding gas mixture.

To overcome the shortcomings of experimental means, numerous scholars have employed numerical simulation to study the effects of shielding gas on physical phenomena and related mechanisms in the welding process. Ogino et al. [23] developed a unified numerical model to explore the effect of Ar and Ar+18%$CO_2$ shielding gas mixtures on the droplet transformation, and the results indicated that the droplet current path at the electrode tip plays a crucial role in determining the transfer mode. Rao et al. [24] investigated the influence of He-Ar gas composition on the GMAW plasma physical properties by establishing a numerical analysis model. The results proved that the shielding gas mixture has an obviously influence on the arc physical properties and droplet transition. Wang et al. [25] utilized a self-developed simulation code to investigate the effects of Ar-$CO_2$ mixture on the temperature distribution and molten pool dynamic behavior in GMAW of steel. The simulation results indicated that with the $CO_2$ content increase, the arc current density and Lorentz force increase, generate a more constricted arc plasma. Consequently, more arc heat reaches the workpiece surface, resulting in a wider and deeper molten pool. However, their models are limited to arc welding applications. Yang et al. [26] established an arc numerical analysis model to study the influence of shielding gas flow rate on MIG and laser-arc hybrid welding. The result found that a high shielding gas flow rate could enhance the droplet transfer stability and, MIG has less process stability than hybrid welding.

From the studies mentioned above, it is evident that the mixed shielding gas has a significant impact on welding quality. However, the influence mechanism of the shielding gas composition on the hybrid plasma physical properties is currently unclear. In the present study, an arc numerical analysis model is established, taking into account the geometric features of the joint and the physical parameters of He-Ar shielding gas. The reasonableness of model is verified through high-speed camera experience. The effects of different He-Ar shielding gas mixture ratios on arc plasma temperature, velocity, and pressure are investigated, respectively. Additionally, the mechanisms of shielding gas composition on the arc plasma physical properties and dynamic characteristics are discussed, which is of great significance to optimize the welding process parameters and improve the stability of the welding process.

## 2. Experimental procedures

Experimental material used is 6061 aluminum alloy and the filler metal is ER5356 aluminum alloy welding wire with a diameter of 1.2 mm, the chemical compositions are shown in Table 1. Fig. 1 shows the schematic diagram of rotating laser-MIG hybrid fillet welding system. The hybrid welding system is composed of an IPG YLS-6000 fiber laser and a Fronius TPS4000 automated welding machine, which is automated using an ABB 6-axis robot. The continuous wave fiber laser with a wavelength of 1075 nm and maximum power of 6 kW, the focal length of 300 mm and the focal spot diameter of 0.3 mm. The welding was carried out with laser-leading and laser-arc distance is 2 mm, and the angle between the laser and the GMAW torch was 30°, other specific dimensions and parameters as displayed in Fig. 1(a) and (b). The process parameters of rotating laser-MIG hybrid welding are listed in Table 2. The shielding gas mixture ratios are 100%Ar, 70%Ar+30%He, 50%Ar+50%He, respectively. The regulation of different helium-argon shielding gas components is achieved by adjusting the gas flow meter on the gas reducer.

Table 1. Composition of welding materials (mass fraction /%).

| **Materials** | **Si** | **Fe** | **Mg** | **Cu** | **Mn** | **Cr** | **Zn** | **Ti** | **Al** |
|---|---|---|---|---|---|---|---|---|---|
| 6061 | 0.4–0.8 | ≤0.7 | 0.8–1.2 | 0.15–0.4 | 0.15 | 0.04~0.35 | 0.25 | ≤0.15 | Bal. |
| ER5356 | ≤0.25 | ≤0.4 | 4.5–5.5 | ≤0.1 | 0.05–0.2 | 0.05–0.2 | ≤0.1 | 0.06–0.2 | Bal. |

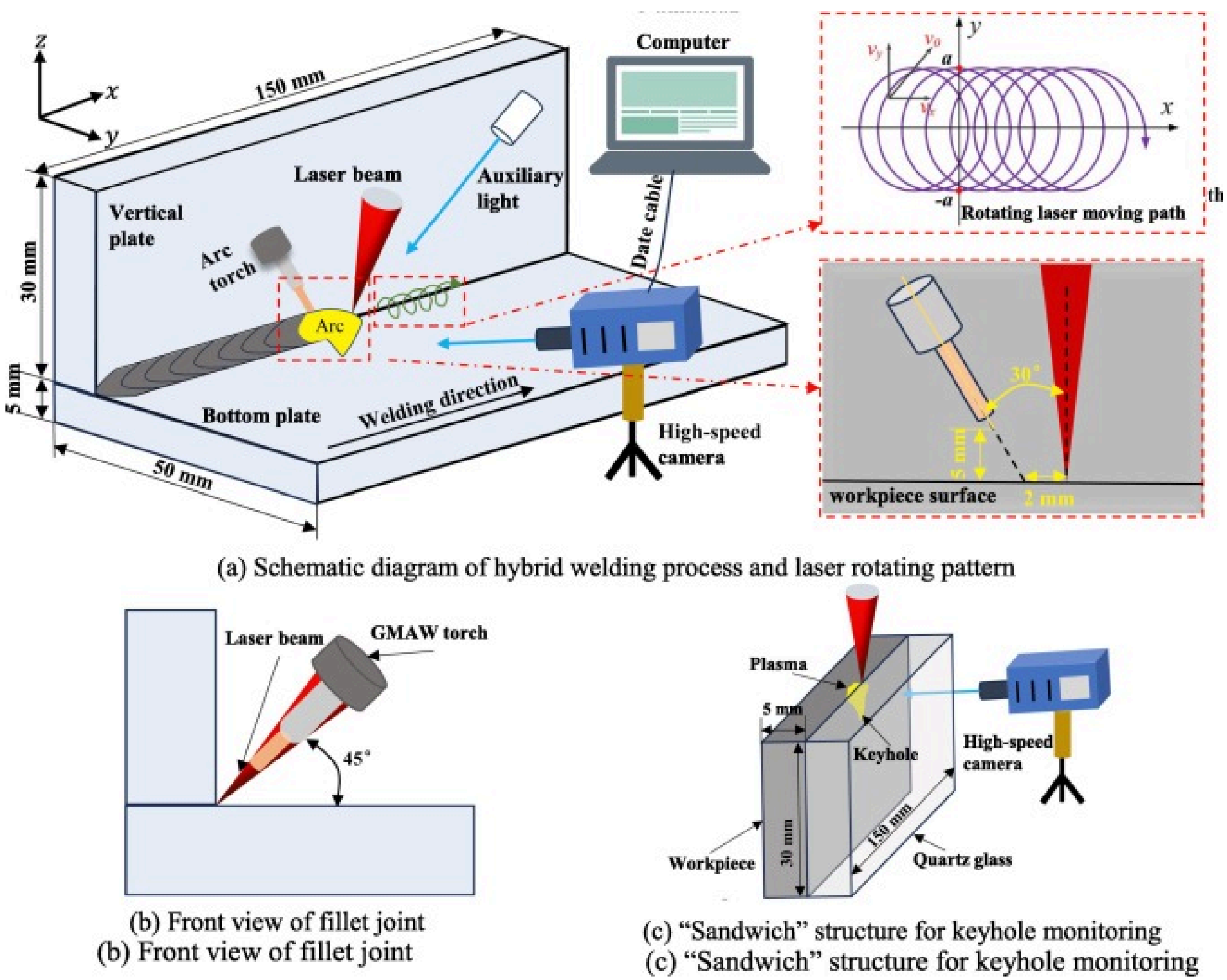




Fig. 1. Rotating laser-MIG hybrid fillet welding system. (a) Schematic diagram of hybrid welding process and laser rotating pattern. (b) Front view of fillet joint. (c) "Sandwich" structure for keyhole monitoring.

Table 2. Experimental parameters of laser+GMAW hybrid welding.

| Process parameters | Value |
|---|---|
| Laser power $P_l$ (kW) | 3 |
| Welding speed $V$ ($m{\cdot}min^{-1}$) | 0.6 |
| Welding current $I$ (A) | 160 |
| Arc and laser angle θ (°) | 30 |
| Defocus amount $\Delta f$ (mm) | 0 |
| Laser rotation radius $a$ (mm) | 1 |
| laser-arc distance $D_l$ (mm) | 2 |
| Rotating frequency $f$ (Hz) | 100 |
| Gas flow $Q_{gas}$ (L/min) | 20 |
| Wire extension (mm) | 15 |

Meanwhile, a high-speed camera system was positioned to record the arc behavior and keyhole size, which includes an auxiliary light source and a high-speed CP80-3-M540 camera, as shown in Fig. 1(a). During welding, the position of hybrid welding torch and high-speed camera are fixed, and data acquisition is realized by moving the stepper control welding platform. The dimensions of the laser-induced keyhole are observed through a “sandwich” structure, which consists of an aluminum alloy workpiece and quartz glass as shown in Fig. 1(c).

## 3. Numerical modeling

### 3.1. Basic assumptions

The hybrid welding process is extremely complex since the arc involves the coupling of multiple physical phenomena such as heat, force, electricity, and magnetism. Coupled with the laser-induced plume (laser-induced plasma and metal vapor) the stirring effect on the plasma also further increases the complexity of the welding process. In order to simplify the calculation and highlight the focus of the study, it is necessary to make a reasonable simplification of the model.

(1) The arc is a laminar flow and incompressible ideal Newtonian fluid;

(2) The arc is in a local thermodynamic equilibrium (LTE);

(3) The metal vapor generated in molten pool、the deformation of molten pool surface and droplet are not considered;

(4) The laser-induced plume is simplified as a metal vapor ejected at high speed from a keyhole.

## 3.2. Governing equations [27]

Based on the assumptions above, the governing equations involved in the numerical analysis are as follows:

Mass

$$\nabla \cdot (\rho v) + \frac{\partial \rho}{\partial t} = 0 \tag{1}$$

where $\rho$ is the density, $v$ is velocity vector.

Momentum

$$\frac{\partial \rho v_x}{\partial t} + \frac{\partial}{\partial x}(\rho v v_x) = -\frac{\partial P}{\partial x} + \frac{\partial}{\partial x}\left[\left(2\mu\frac{\partial v_x}{\partial x}\right) - \frac{2}{3}\nabla \cdot v\right] + \frac{\partial}{\partial y}\left[\mu\left(\frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x}\right)\right] + \frac{\partial}{\partial z}\left[\mu\left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x}\right)\right] + j_y B_z - j_z B_y \tag{2}$$

$$\frac{\partial \rho v_y}{\partial t} + \frac{\partial}{\partial y}(\rho v v_y) = -\frac{\partial P}{\partial y} + \frac{\partial}{\partial y}\left[\left(2\mu \frac{\partial v_y}{\partial y}\right) - \frac{2}{3}\nabla \cdot v\right] + \frac{\partial}{\partial z}\left[\mu\left(\frac{\partial v_y}{\partial z} + \frac{\partial v_z}{\partial y}\right)\right] + \frac{\partial}{\partial z}\left[\mu\left(\frac{\partial v_y}{\partial x} + \frac{\partial v_x}{\partial y}\right)\right] + j_z B_x - j_x B_z \tag{3}$$

$$\frac{\partial \rho v_z}{\partial t} + \frac{\partial}{\partial y}(\rho v v_z) = -\frac{\partial P}{\partial z} + \frac{\partial}{\partial z}\left[\left(2\mu \frac{\partial v_y}{\partial y}\right) - \frac{2}{3}\nabla \cdot v\right] + \frac{\partial}{\partial x}\left[\mu\left(\frac{\partial v_z}{\partial x} + \frac{\partial v_x}{\partial z}\right)\right] + \frac{\partial}{\partial y}\left[\mu\left(\frac{\partial v_z}{\partial y} + \frac{\partial v_y}{\partial z}\right)\right] + j_x B_y - j_y B_x \tag{4}$$

where $v_x$, $v_y$ and $v_z$ are the velocity vectors in the$x$,$y$ and $z$ direction, respectively; $P$ is the pressure;$\mu$is the viscosity coefficient; $j_x$, $j_y$ and $j_z$ are the current density components in the three directions, respectively; $B_x$, $B_y$ and $B_z$ are the induced magnetic field intensity components in three directions, respectively.

Energy

$$\frac{\partial \rho h_e}{\partial t} + \nabla \cdot (\rho v h_e) = \nabla \cdot \left(\frac{k}{c_p}\nabla h_e\right) + \frac{j_x^2 + j_y^2 + j_z^2}{\sigma_e} - 4\pi\zeta + \frac{5k_B}{2e} j \cdot \nabla T + Q_{laser} \tag{5}$$

where $c_p$ is the specific heat capacity; $h_e$ is the enthalpy; $\sigma_e$ is the electrical conductivity; $k_B$ is the Boltzmamn constant; $\zeta$ is the net radiation coefficient; The second term, third term and fourth term on the right side of the equation are the Joule heat, the heat lost to the outside world, and the heat of electron migration, respectively; $Q_{laser}$ denotes the energy of the laser absorbed by the arc.

Electromagnetic field equation

Current continuity equation

$$\frac{\partial}{\partial x}\left(\sigma_e \frac{\partial \phi}{\partial x}\right) + \frac{\partial}{\partial y}\left(\sigma_e \frac{\partial \phi}{\partial y}\right) + \frac{\partial}{\partial z}\left(\sigma_e \frac{\partial \phi}{\partial z}\right) = 0 \tag{6}$$

Current density component

$$j_x = -\sigma_e \frac{\partial \phi}{\partial x}, j_y = -\sigma_e \frac{\partial \phi}{\partial y}, j_z = -\sigma_e \frac{\partial \phi}{\partial z} \tag{7}$$

where $\phi$ is the electric potential.

Magnetic field components

$$\frac{\partial^2 A_x}{\partial x^2}+\frac{\partial^2 A_x}{\partial y^2}+\frac{\partial^2 A_x}{\partial z^2}=-\mu_0 j_x \tag{8}$$

$$\frac{\partial^2 A_y}{\partial y^2}+\frac{\partial^2 A_y}{\partial x^2}+\frac{\partial^2 A_y}{\partial z^2}=-\mu_0 j_y \tag{9}$$

$$\frac{\partial^2 A_z}{\partial z^2}+\frac{\partial^2 A_z}{\partial y^2}+\frac{\partial^2 A_z}{\partial x^2}=-\mu_0 j_z \tag{10}$$

$$B_x=\frac{\partial A_z}{\partial y}-\frac{\partial A_y}{\partial z}, B_y=\frac{\partial A_x}{\partial z}-\frac{\partial A_z}{\partial x}, B_z=\frac{\partial A_y}{\partial x}-\frac{\partial A_x}{\partial y} \tag{11}$$

where $A_x$, $A_y$ and $A_z$ are the magnetic vector components in the $x$, $y$ and $z$ directions, respectively; $\mu_0$ is the vacuum permeability.

Metal vapor transport equation

$$\frac{\partial}{\partial t}(\rho f_v)+\nabla\cdot(\rho v f_v)=\nabla\cdot(\rho D\nabla f_v) \tag{12}$$

where $f_v$ is the mass fraction of metal vapor; $D$ is the diffusion coefficient, which is determined by the viscosity approximation equation.

$$D=\frac{2\sqrt{2}(1/m_1+1/m_2)^{0.5}}{\left\{\left(\rho_1^2/\beta_1^2\eta_1^2 m_1\right)^{0.25}+\left(\rho_2^2/\beta_2^2\eta_2^2 m_2\right)^{0.25}\right\}^2} \tag{13}$$

where $m_1$ and $m_2$ are the molar masses of aluminum and mixed shielding gases, respectively; $\rho_1$, $\rho_2$, $\eta_1$, and $\eta_2$ are the densities and viscosity of aluminum metal vapor and shielding gases, respectively; $\beta_1$ and $\beta_2$ are dimensionless constants, and the theoretical values of common shielding gases Ar, He, and $O_2$ range from 1.2 to 1.543 [28]. Based on the experimental data in reference [29], this article assumes that $\beta_1$=$\beta_2$=1.385.

The rotating laser affects the magnitude of the arc temperature, the mass, and the energy transfer; Hence, a reasonable and accurate description of the laser rotation path is required. rotating path formula [30].

$$\begin{cases} x(t) = x_0 + a \cdot \cos(2\pi \cdot f \cdot t) \\ y(t) = a \cdot \sin(2\pi \cdot f \cdot t) \end{cases} \tag{14}$$

where $a$ is the radius of the laser rotation, $f$ is the frequency, and $x_0$ is the position of the laser rotation centre in the x-axis. The rotating laser moving path is shown in Fig. 1(a).

### 3.3. Laser induced metal vapor model

In the numerical analysis model of hybrid welding, it is also very important to obtain the morphology and radius of the high-speed laser-induced plasma and the keyhole. In this study, the morphology of keyhole under different laser power is obtained by a high-speed camera as shown in Fig. 2. As the laser power is enhanced, the size of keyhole gradually increases.

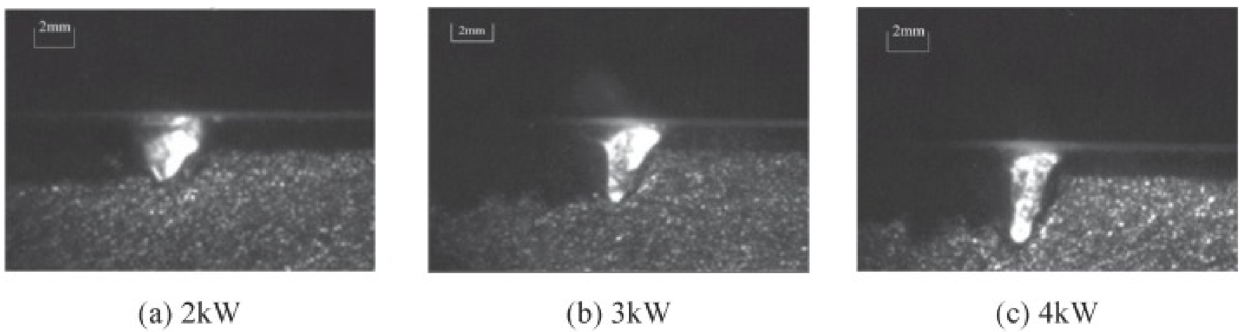


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Fig. 2. Keyhole morphology corresponding to different laser power. (a) 2kW. (b) 3kW. (c) 4kW.

The relationship between the keyhole radius, plasma temperature and laser energy density can be described by the keyhole laser-induced plasma model established by

Dilthey [31].

$$T = 0.00425 \times I_l + 3240.38462, 0.20\text{mm} < R \leq 0.30\text{mm} \tag{15}$$

$$T = 0.00431 \times I_l + 3252.13675, 0.30\text{mm} < R \leq 0.40\text{mm} \tag{16}$$

$$T = 0.00438 \times I_l + 3263.88889, 0.40\text{mm} < R \leq 0.50\text{mm} \tag{17}$$

where $T$ is the plasma temperature; $I_l$ is the laser energy density; $R$ is the radius of keyhole.

The velocity of metal vapor at the keyhole was calculated according to the model developed by Amara [32].

$$V_g = \sqrt{\kappa \frac{LT_k}{m_a}} \tag{18}$$

where $\kappa$ is the specific heat ratio; $L$ is the ideal gas constant; $T_\text{k}$ is the surface temperature of laser keyhole; $m_\text{a}$ is the atomic mass.

### 3.4. Boundary conditions

The boundary conditions of the model are listed in Table 3. In the table, $r$ is the distance from the center of rotating laser moving path, and $R$ is the eruption radius of keyhole on the workpiece surface. The boundary conditions of the laser-induced metal vapor model are as follows: if $r \leq R$, the laser-induced metal vapor is located in the keyhole region, the mass fraction of metal vapor is 100%, the plasma eruption velocity is $V_g$, and the temperature is $T_g$; if $r>R$, which is outside the keyhole range, the aluminum vapor eruption velocity is 0K and the temperature is 1000K. In order to ensure that the helium-argon mixed shielding gas ionization process proceeds within the arc area, the initial temperature of calculation area is set to 8000K. To simplify the model, the current density at the electrode tip is homogeneous model.

$$J_Z = \frac{I^2}{\pi r\ w} \tag{19}$$

where $I$ is the welding current and $r_w$ is the wire radius.

Table 3. Boundary conditions.

| **Boundary** | ***F*** | ***V*(m·s$^{-1}$)** | ***T*(K)** | ***φ*(V)** | ***A*(Wb·m$^{-1}$)** |
|---|---|---|---|---|---|
| Inlet | | $V_a$ | 1000 | $\frac{\partial\varphi}{\partial n} = 0$ | $\frac{\partial A}{\partial n} = 0$ |
| Outlet | 0 | $\frac{\partial V}{\partial n} = 0$ | 1000 | $\frac{\partial\varphi}{\partial n} = 0$ | 0 |
| Wire tip | 0 | 0 | 3000 | $J_Z$ | $\frac{\partial A}{\partial n} = 0$ |
| Wire wall | 0 | 0 | 1000 | $\frac{\partial\varphi}{\partial n} = 0$ | $\frac{\partial A}{\partial n} = 0$ |
| Workpiece | 1,$r$ <=$R$<br>0,$r$>$R$ | $V_g$, $r$ <=$R$<br>0,$r$>$R$ | $T_g$, $r$ <=$R$<br>1000,$r$>$R$ | 0 | $\frac{\partial A}{\partial n} = 0$ |

The hybrid welding arc computational domain is as illustrated in Fig. 3 (a). The length of calculation area in x-direction is 25 mm and in z-direction is 7 mm. The initial arc length is 5 mm, the distance from the upper surface is 2 mm. The laser-arc distance varies periodically as the laser rotates. The schematic diagram of dividing the model boundary conditions is shown in Fig. 3 (b). The welding wire end is set as the anode, while the rest of its face is set as the wall. The lower surface of the workpiece is set as the cathode. The position of keyhole is set as the velocity inlet for laser-induced metal vapor eruption. The upper surface of the model is set as the shielding gas inlet, and the remaining boundary is set as the pressure outlet.

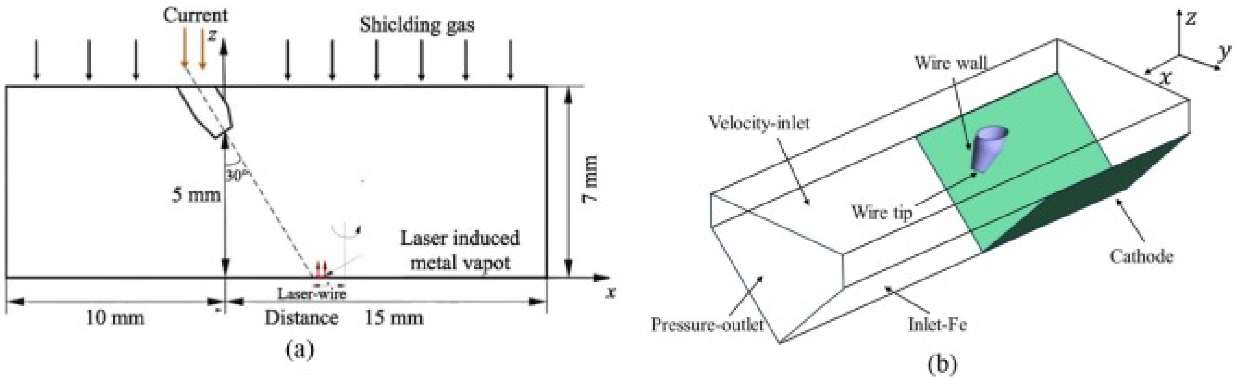


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Fig. 3. Schematic diagram of model: (a) calculation domain; (b) boundary conditions.

In this study, the non-uniform mesh is used to divide the model, a denser grid was used in the near weld wire and laser induced metal vapor eruption region with a minimum size of 0.1 mm as shown in Fig. 4. The remaining regions use a sparse grid with a maximum size of 0.4mm. The model has a total of 239,705 grid nodes. In addition, the non-uniform time step is used and the minimum time step of $10^{-6}$s.

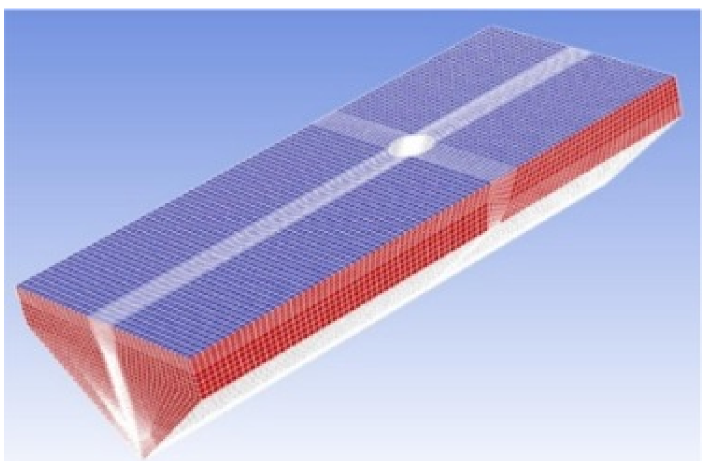

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Fig. 4. Schematic diagram of grid for arc model.

### 3.5. Temperature-dependent properties of shielding gas

Physical property parameters of metallic aluminum, argon and helium vary with temperature during high-temperature welding, which affect the accuracy of final calculations. The partial thermal-physical property parameters of pure Ar, pure He, and He-Ar mixtures are given in equilibrium state are drawn in Fig. 5 [33], [34], [35], [36]. The remaining physical property parameters of helium-argon can be obtained by linear interpolation.

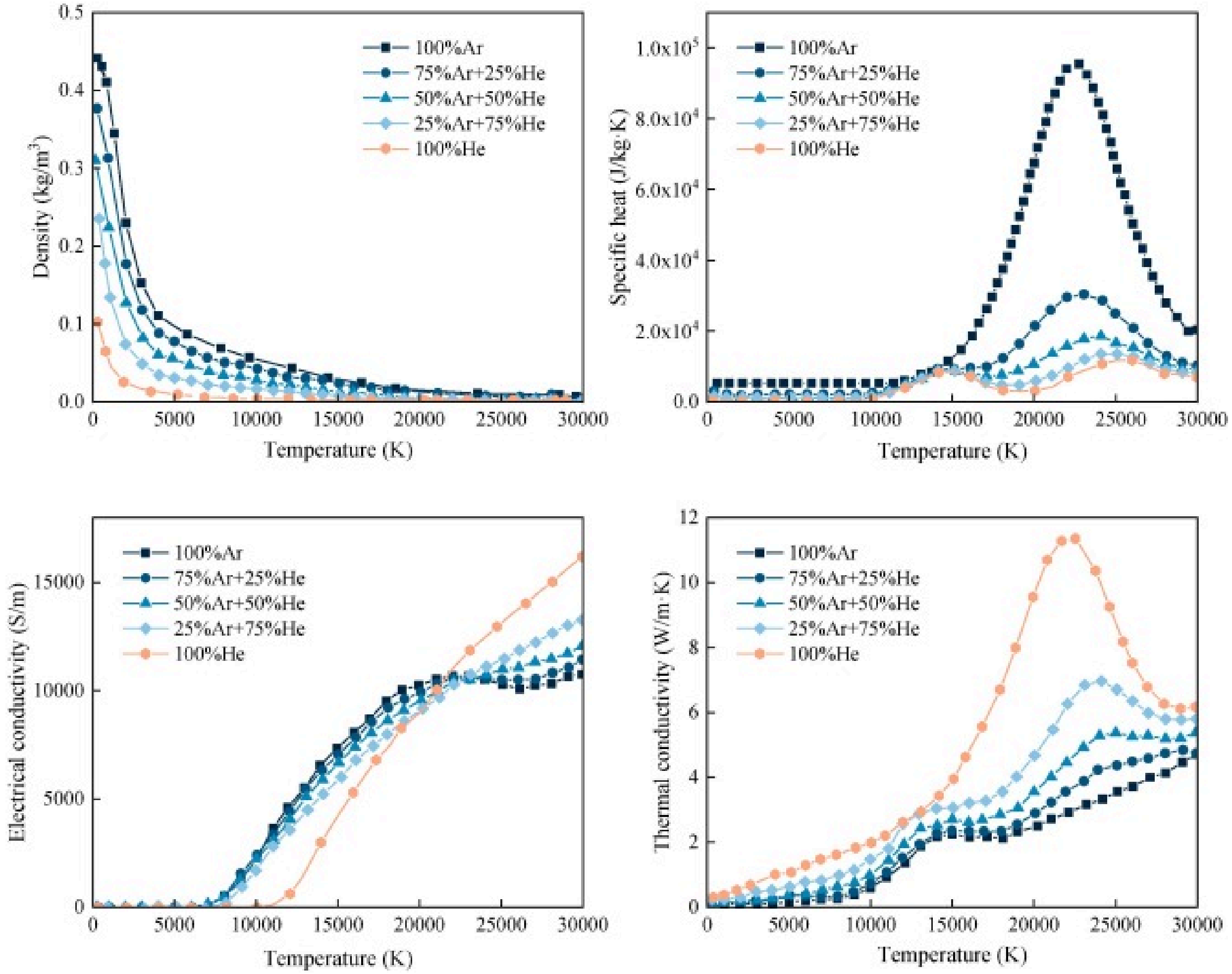


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Fig. 5. Temperature-dependent material properties of mixed shielding gases.

## 4. Results and discussion

The direction and morphology of laser-induced metal plume changes periodically due to the rotation of laser, which affects the physical properties of arc plasma at various locations. For this reason, in this study, plasma distribution cloud diagrams of different sections with the laser at various positions are selected for analysis. The selected longitudinal sections are S1, S2, and S3, the three sections are parallel to each other, and the S2 and S3 sections are the positions where laser rotates to the vertical plate (D) and bottom plate (B), as shown in Fig. 6. The S4 cross-section is the position of laser beam center, and the rotation direction is A-B-C-D-A. A and C positions are located on the weld center axis and are the furthest and closest to the wire, respectively. B and D are located on both sides of the weld at a distance of 1 mm from the weld center.

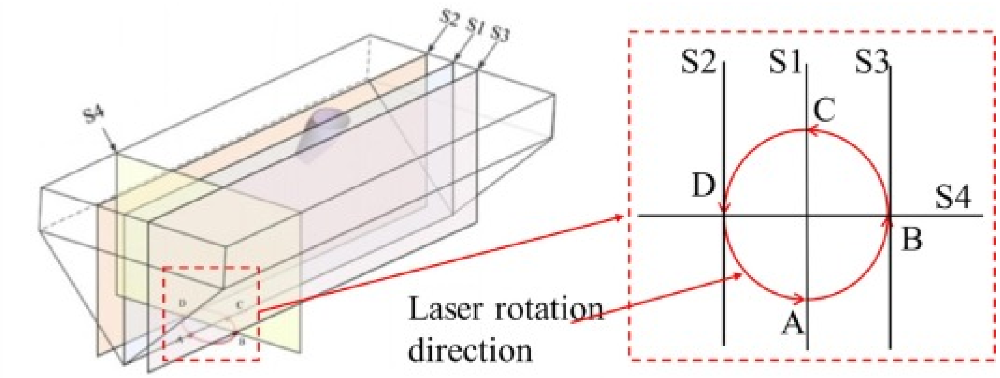


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Fig. 6. Schematic diagram of laser position and rotation direction.

### 4.1. Temperature distribution

Fig. 7 shows the distributions of arc plasma longitudinal section temperature field at position A under varying shielding gas compositions. The peak temperature of arc plasma occurs near the electrode tip and decreases with increasing distance from the electrode tip. With the increase of He content in shielding gas, the overall shape of arc plasma contracts and move towards the laser side in the three longitudinal sections (S1, S2, S3). The temperature distribution range on the left side of the arc significantly decreases when the He gas content is 30%. While the He content reaches 50%, the distribution range on the right side of the arc significantly compresses, the overall temperature distribution of arc is more concentrated, as depicted in Fig. 7 (a1-a3). Meanwhile, the arc peak temperatures in S1 plane are 24081.8 K, 20949.8 K and 19760.1 K when the He content ranges from 0% to 50%, respectively. As the He content increases, the peak temperature of plasma significantly decreases in the initial stage, and then the downward trend slows down. Cai [16] obtained a similar trend of arc temperature and size variation by a spectral acquisition system and a high-speed video. The above phenomena are associated with the ionization potential and thermal conductivity of He, which will be further discussed below.

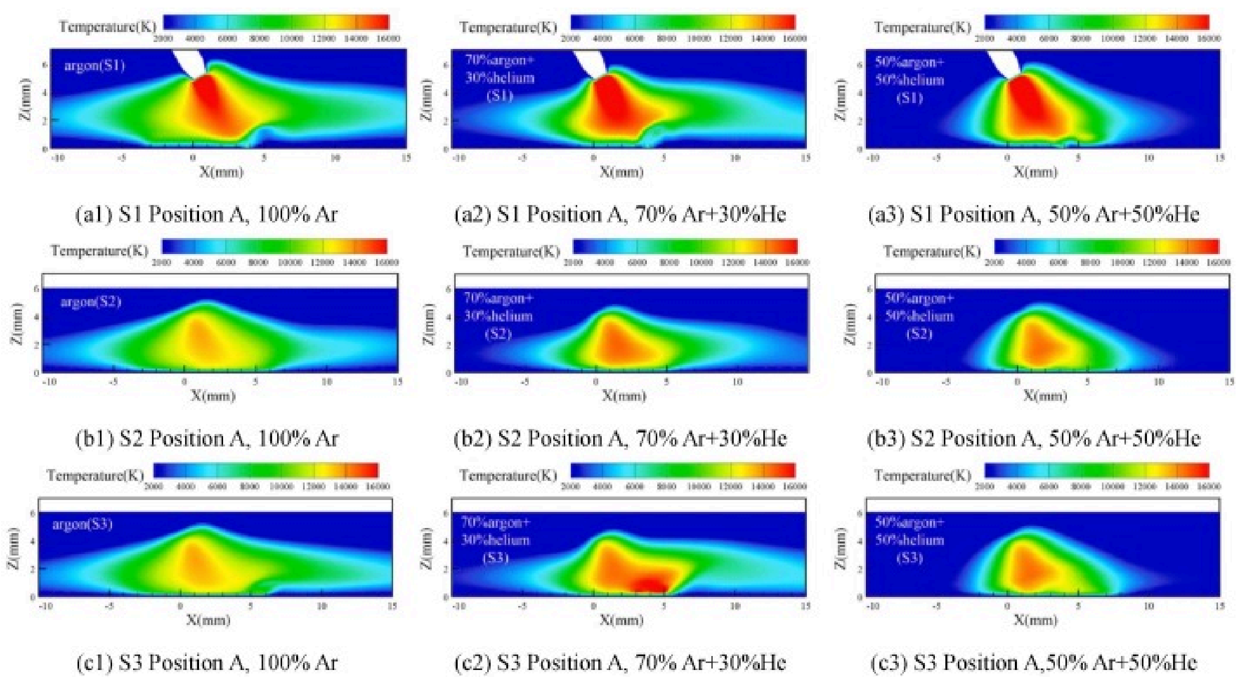


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Fig. 7. Longitudinal section temperature distributions at A positions.

The longitudinal sections S2 and S3 are symmetric about the center of weld, but the arc plasma is less affected by the metal vapor and the plasma is less dispersed on the right side of arc in the S2 section than in the S3 section. This is since the former is located in the vertical plate, where the kinetic energy of laser-induced metal vapor is contracted to the left side of laser under the influence of rotational tangential force. While the latter is located in the horizontal plate, the plasma moves to the right side of laser under the counterclockwise rotational tangential force, as shown in Fig. 7 (b1) and (c1). Rotating laser can improve the arc distribution characteristics in the fillet joint, avoid arc burning on a single workpiece plate, and suppress undercut defects caused by insufficient melting of the vertical plate. The S1 section is located in the center of weld seam, the laser-induced metal vapor and the arc plasma are on the same axis, where both meeting heights are higher compared to the S2 and S3 sections, as shown in Fig. 7 (a1). In addition, compared to the S2 and S3 section, the S1 section has a larger high-temperature region in the middle of electrode and keyhole, due to the S1 section is closer to the electrical tip. The arc contraction effect of increasing He content in the shielding gas on longitudinal sections S2 and S3 is similar to that on section S1, which will not be repeated here, and the following analyses also focus on the arc physical properties in the S1 longitudinal section as the laser is rotated to different positions.

Fig. 8 displays the temperature distributions of arc longitudinal section(S1) under varying shielding gas compositions. The laser beam moves continuously according to the rotation path (as shown in Fig. 6), the disturbed position of arc plasma by the laser is constantly changing, and the arc shape of hybrid welding is always in the process of dynamic change. The arc overall morphology at positions B, C, and D follows the variation trend observed at position A with the He gas content increases in the shielding gas. Since positions B and D are located on both sides of the weld and are symmetrical with respect to the weld, the arc profiles of B and D show a similar trend as the proportion of He increases. However, position C is closer to the electrical tip than A, so the plasma is affected by the laser-induced metal vapor at roughly $X = 2$ mm, while position A is at $X = 4.5$ mm. In this paper, the laser rotation radius is set to 1 mm, and the distance between the constrained positions of A and C conforms to the model settings, which explain indirectly the correctness of model.

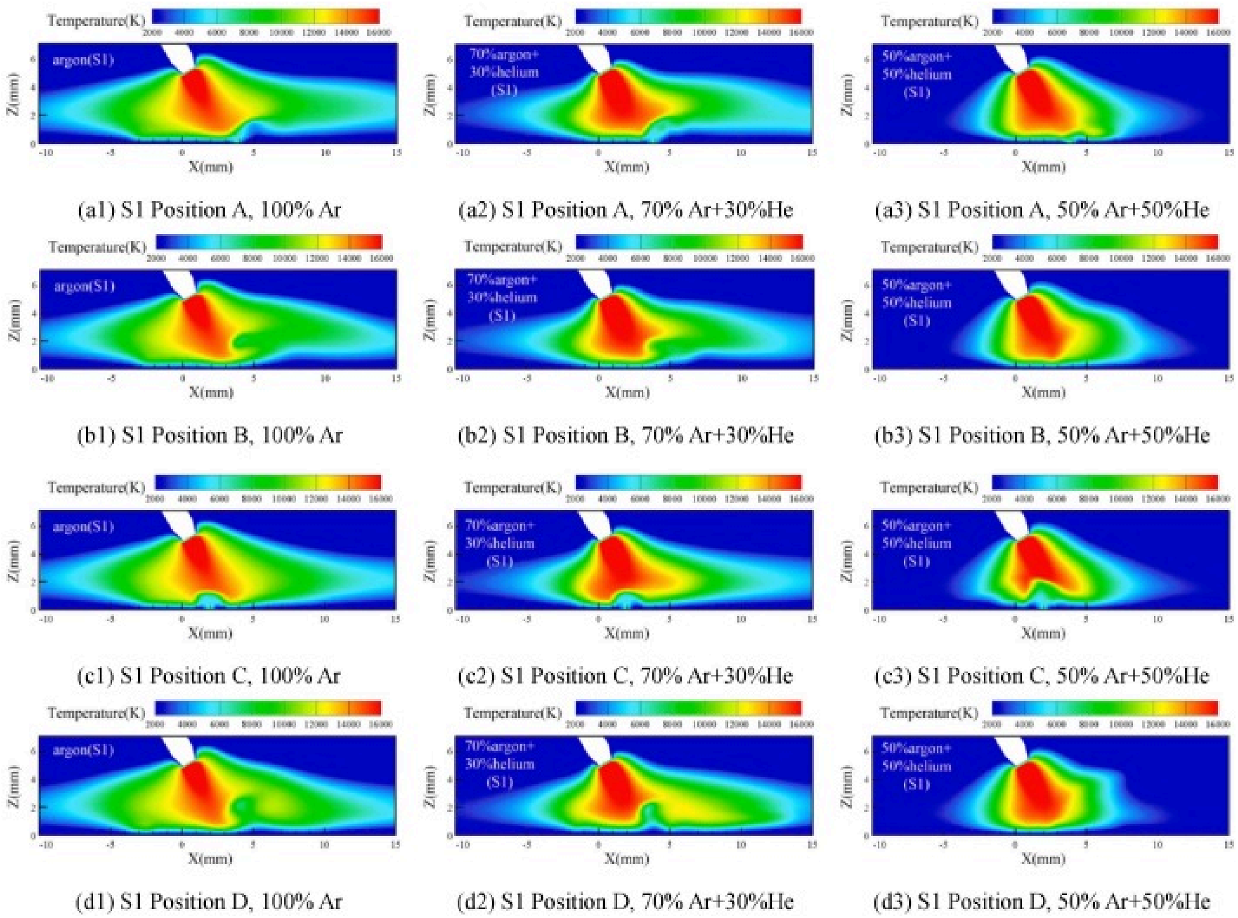


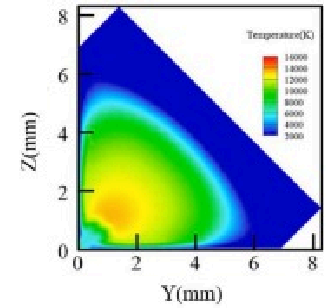



Fig. 8. Longitudinal section temperature distributions at various positions.

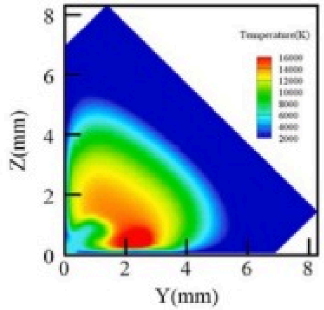
The distribution range and temperature of arc plasma decrease significantly with the increase of He content, which is ascribed to the higher ionization potential of helium, the electric conductivity sharply decreases with the increase of He content at the arc temperature is below 20000K, as shown in Fig. 5. During the welding process, compared to pure argon, the ionization difficulty of He-Ar mixed shielding gas is higher, and the plasma size formed by it becomes smaller, leading to a significant arc contraction [16],

[26]. After the arc contraction, the shielding effect of plasma on the laser is weakened, which is beneficial for improving the effective energy density of the laser, and higher laser energy density also brings greater welding penetration depth [37]. According to the principle of minimum arc pressure, the arc will be easier to choose keyhole as the conductive channel, and the arc size will be further reduced. The high thermal conductivity of He is another crucial factor influencing energy absorption during the welding process. With the increase of He content, the thermal conductivity of mixed shielding gas increases sharply, as shown in Fig. 5, which accelerates the heat transfer rate between the plasma and the surrounding air and increases the cooling rate of plasma [38], [39]. It is easy to exchange heat with the surrounding environment through thermal convection and thermal radiation, has a cooling effect on the arc, and maintains energy balance by shrinking the arc. Meanwhile, rotating laser can also accelerate the decrease in arc plasma temperature [40]. A suitable He-Ar shielding gas mixture can compensate for the shortcomings of rotating laser-arc hybrid welding itself.

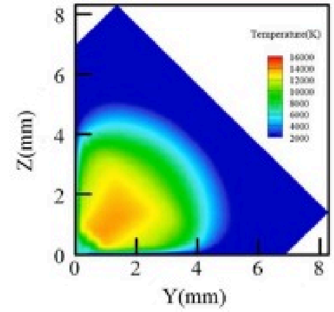
Fig. 9 shows the temperature distributions of arc cross-sectional (S4) under varying shielding gas compositions. When the laser is located in the B and D position, due to the rotation of laser heat source, the rotational tangential forces experienced by both are in different directions, causing the arc plasma high-temperature region to deflect in opposite directions, as shown in Fig. 9(b1) and (d1). The overall morphology of arc plasma at positions A and C is similar and distributed symmetrically about the center of weld, due to position C is closer to the electrical tip, the arc high-temperature region in the C is obviously larger than in the position A. When the proportion of He is 30% in the shielding gas, the high-temperature region at positions A and C is shifted towards the horizontal plate. This is due to the fact that with the addition of He, the shielding effect of hybrid plasma on the laser is weakened, the effective laser density is increased, more metal is melted, and the liquid metal is deflected towards the horizontal plate by gravity. As the He content reaches 50%, the energy distribution of the arc becomes more concentrated, and the high-temperature shift of arc at positions A and C also disappears. From the cross-section of arc, for low He content, the distribution of arc plasma is affected by the rotating laser and deviates at various positions. As the He content increases, the mutual influence between laser and arc plasma is weakened, as described above., which is more conducive to the arc stability.



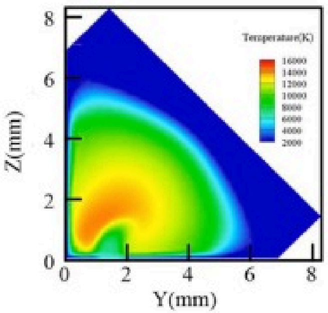

(a1) S4 Position A,100%Ar



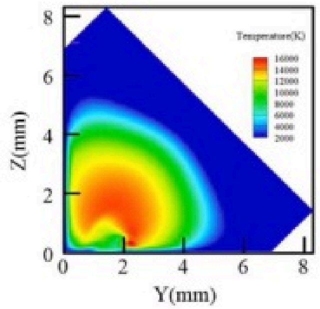

(a2) S4 Position A,70%Ar+30%He



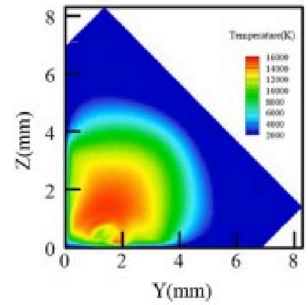

(a3) S4 Position A,50%Ar+50%He



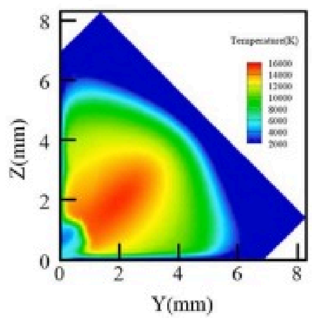

(b1) S4 Position B,100Ar



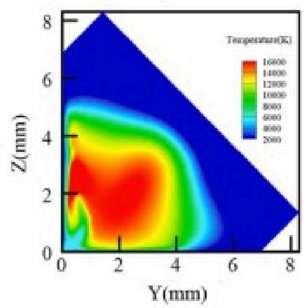

(b2) S4 Position B,70%Ar+30%He



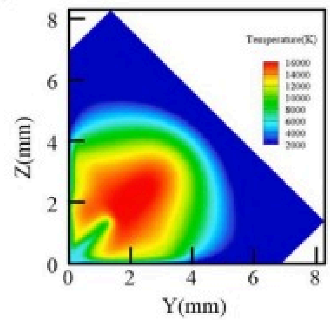

(b3) S4 Position B,50%Ar+50%He



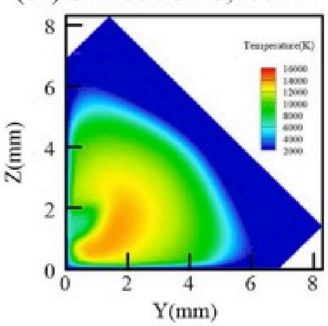

(c1) S4 Position C,100Ar



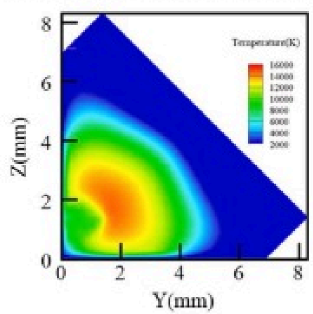

(c2) S4 Position C,70%Ar+30%He



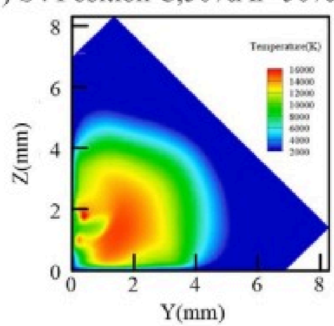

(c3) S4 Position C,50%Ar+50%He




(d1) S4 Position D,100Ar




(d2) S4 Position D,70%Ar+30%He




(d3) S4 Position D,50%Ar+50%He

Fig. 9. Cross-sectional temperature distributions at various positions.

Fig. 10 shows the temperature distribution curves of arc plasma near the workpiece surface at various positions. Compared with the pure argon, when the shielding gases are 70%Ar+30%He and 50%Ar+50%He, the temperature curve to the left of peak temperature is shifted to the positive direction of X-axis by about 2–5 mm, which is due to the fact that the addition of He gas constricts arc plasma, increases the heat flux density of arc, and the high-temperature region of arc is more concentrated, so the curve is shifted to the right. Compared with position A, position C is closer to the welding wire, so the peak temperature zone moves 2 mm to the left relatively, as shown in Fig. 10 (a) and (c). The temperature distribution characterized of arc plasma is higher in the middle and lower on both sides on the surface of workpiece, when the lasers are located at B and D, respectively. When the shielding gas is 50%Ar+50%He, there is a low peak temperature area on the right side of position D. This could be caused by the fact that D is located on the vertical plate, liquid metal moves toward the bottom plate under the action of gravity, and the arc temperature decreases.

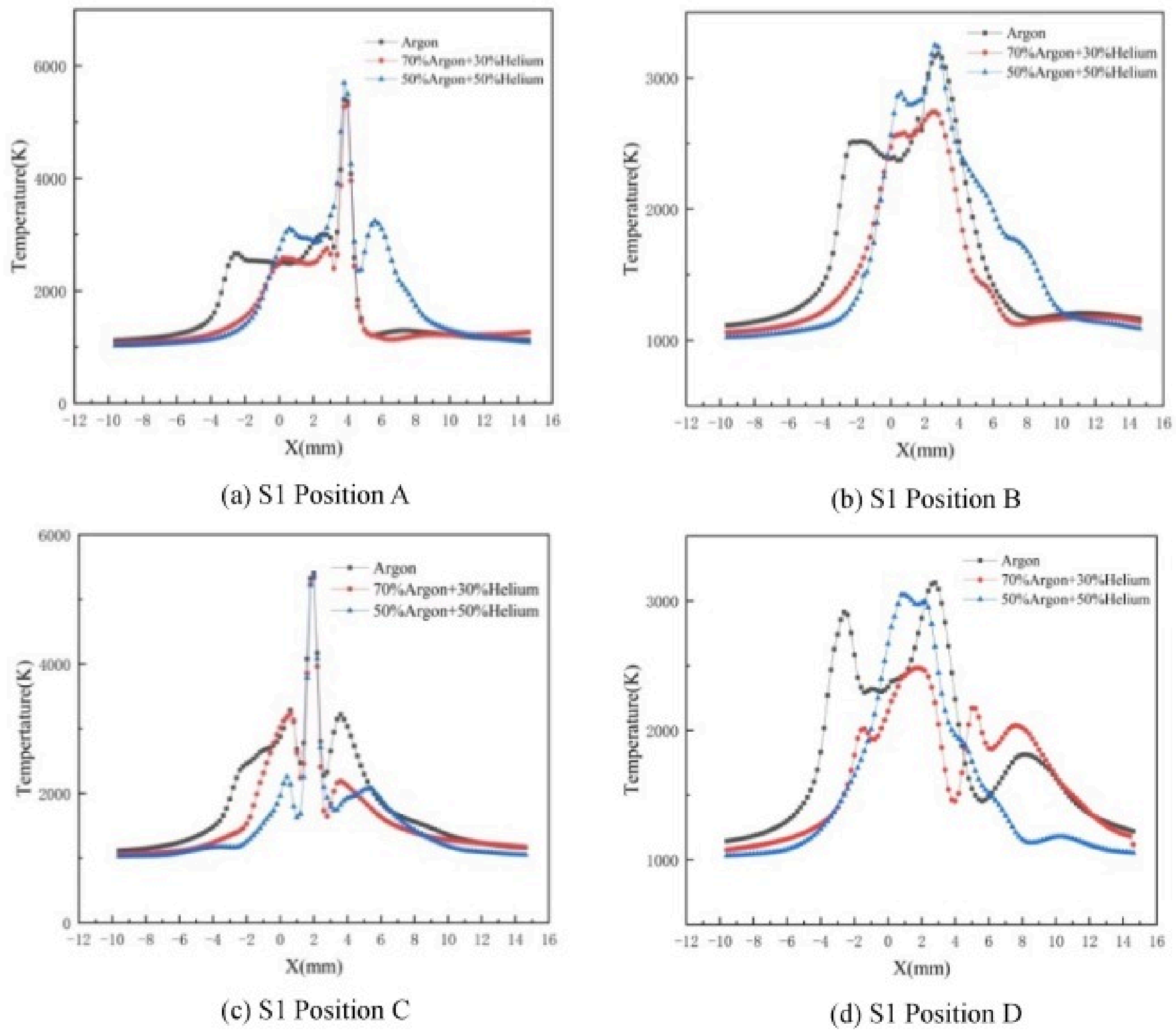



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Fig. 10. Workpiece surface temperature distribution at various positions. (a) S1 Position A. (b) S1 Position B. (c) S1 Position C. (d) S1 Position D.

## 4.2. Plasma velocity

Fig. 11 shows the velocity distributions of arc plasma longitudinal section under varying shielding gas compositions. The arc plasma flows at high speed along the direction of welding torch, after the plasma reaches the surface of workpiece, it flows along the positive and negative directions of x-axis, and the arc plasma is deflected upward by the impact of laser-induced metal plume at the keyhole as shown in Fig. 11(a1), (b1) and (d1). The flow pattern at position C varies depending on the distance between the laser and the wire decreases, as shown in Fig. 11(c1). The highest plasma flow velocity at each position under different shielding gases is located near the electrode tip, which are, respectively, 317.3 m/s, 317.2 m/s, 317.6 m/s, and 316.4 m/s for pure Ar. The peak flow velocity are, respectively, 320.3 m/s, 320.6 m/s, 319.3 m/s and 318.4 m/s for 30% He content. The maximum flow velocity of plasma further increased, which are, respectively, 444.2 m/s, 444.4 m/s, 442.6 m/s, and 441.9 m/s for 50% He content. As the He content increases, the distribution range of arc plasma decreases, and its flow rate increases within a limited area. With the further increase of He content, coupled with the increase of laser induced metal vapor kinetic energy, the plasma flow rate significantly increases.

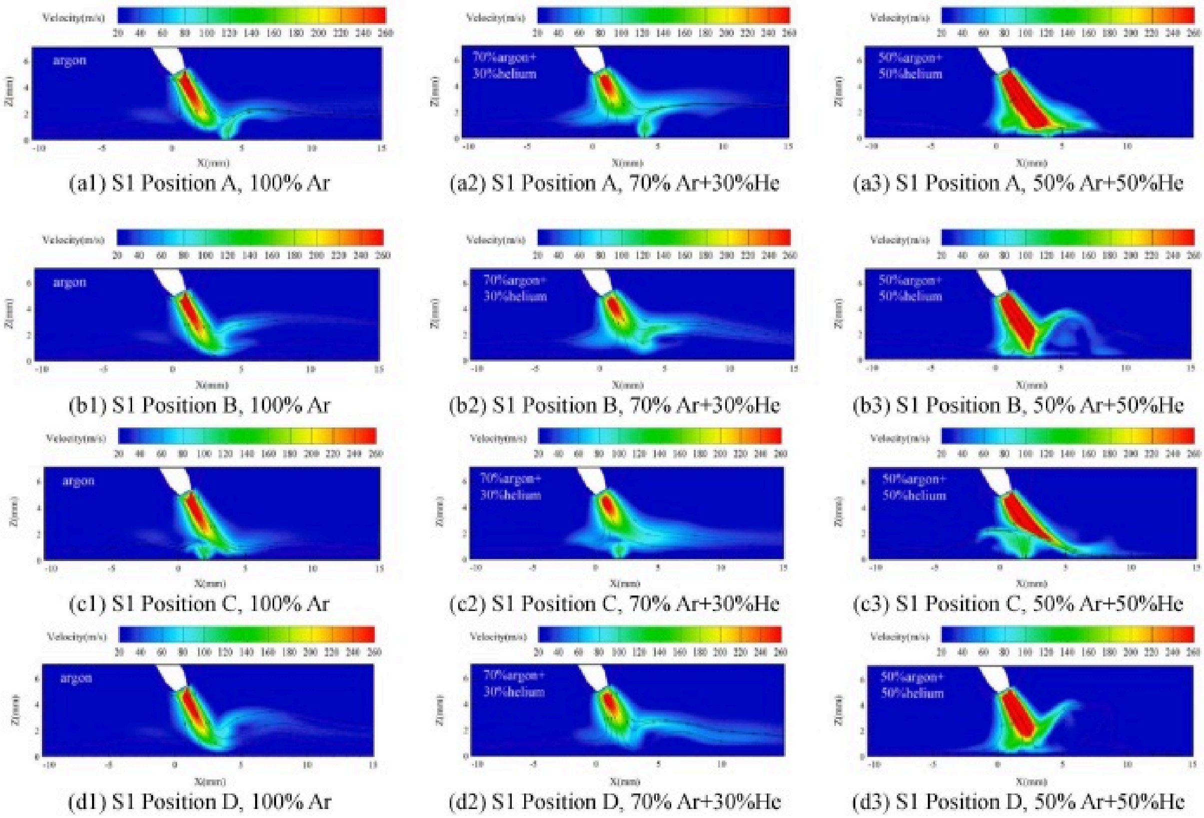


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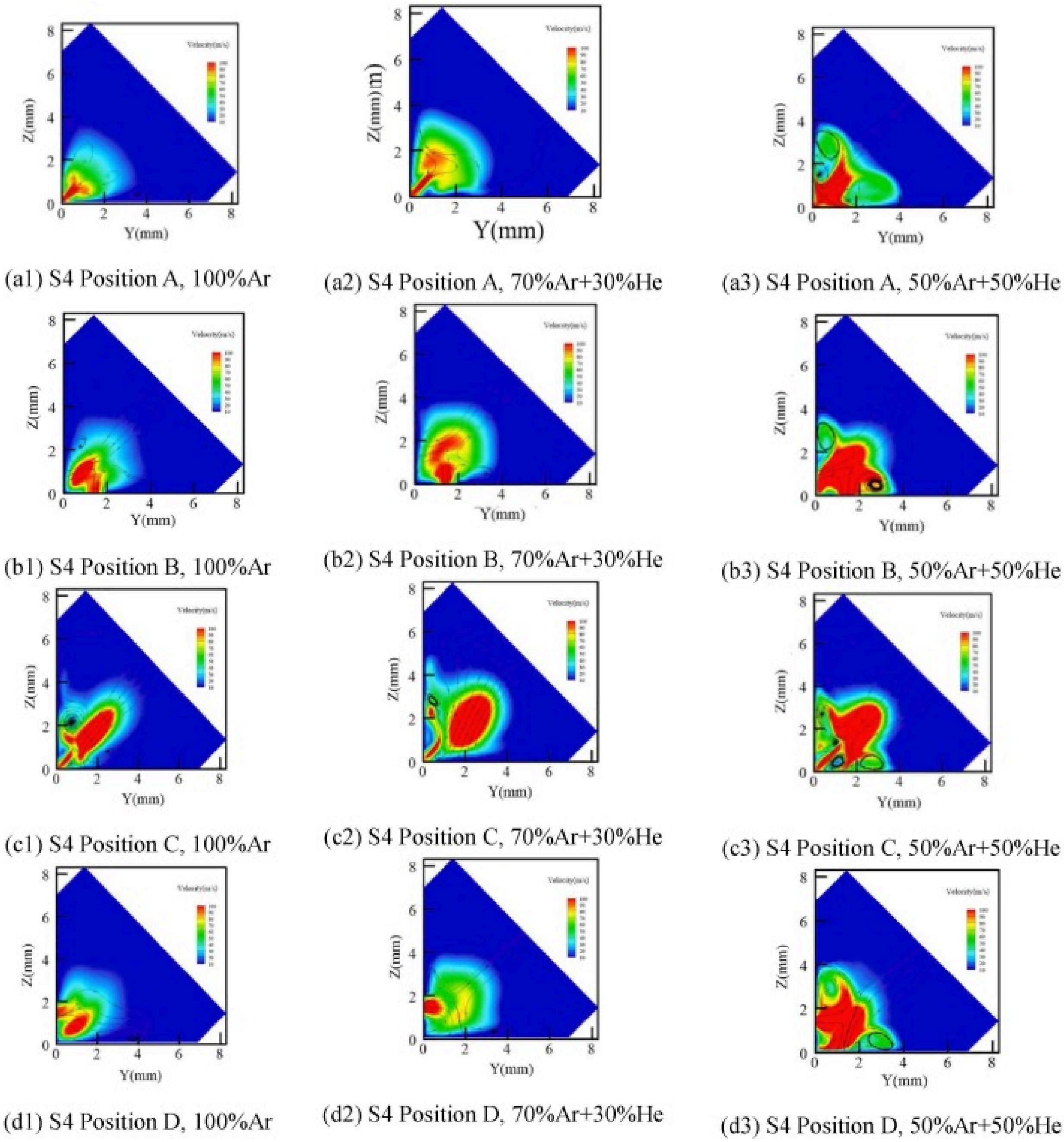
Fig. 11. Longitudinal section plasma flow field at various positions.

Fig. 12 shows the velocity field distributions of arc plasma cross-section (S4) under varying shielding gas compositions. When the shielding gas is Ar, the plasma peak flow velocity at position A is 125.4m/s. When the laser at position C, due to the smallest distance between laser and welding wire, coupled with the influence of tangential force of aluminum metal vapor, it can be seen that the plasma has an obvious vortex phenomenon, and the peak flow velocity increases to 228.3m/s, as shown in Fig. 12(c1). While the laser heat source rotates to positions B and D, the forces they receive are in different directions, and the plasma is deflected to both sides of the workpiece. The peak flow velocities are 141.4 m/s and 128 m/s, respectively, as shown in Fig. 12(b1) and (d1). When the He content is 30%, the maximum flow velocity of plasma is 134.1 m/s, 144.4 m/s and 116 m/s in the cross section of A, B and D positions, respectively, which is increased compared with that of pure Ar. However, the peak plasma flow velocity at the C position decreases to 192.5 m/s and the area of vortex is reduced. When the He content reaches 50%, the plasma flow velocity spreads to both sides of the workpiece, the flow pattern becomes more complex, and vortex phenomenon appear on the both sides of welding wire, the maximum flow velocity increased to 250.3 m/s, 292.9 m/s, 395.8 m/s and 284.6 m/s, respectively, shows large differences. The calculations indicate that when the He content is 50%, the arc plasma is more easily affected by the position of laser heat source and the aluminum metal vapor, and the flow pattern is disordered, which is detrimental to the welding process stability. The inert gas Ar has a good arc starting and stabilizing effect [41], as He increases and Ar decreases in the shielding gas, the stability of arc is suppressed. From the above analysis, it can be observed that there is an optimal range for the He content in the shielding gas. When the proportion of He is too large, it is not conducive to the stability of arc [21]. In addition, when the proportion of He in the shielding gas mixture is high, the C position in rotating laser-arc hybrid welding is prone to flow state instability because of laser-induced metal vapor.




(a1) S4 Position A, 100%Ar (a2) S4 Position A, 70%Ar+30%He (a3) S4 Position A, 50%Ar+50%He

(b1) S4 Position B, 100%Ar (b2) S4 Position B, 70%Ar+30%He (b3) S4 Position B, 50%Ar+50%He

(c1) S4 Position C, 100%Ar (c2) S4 Position C, 70%Ar+30%He (c3) S4 Position C, 50%Ar+50%He

(d1) S4 Position D, 100%Ar (d2) S4 Position D, 70%Ar+30%He (d3) S4 Position D, 50%Ar+50%He

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Fig. 12. Cross-sectional plasma flow field at various positions.

### 4.3. Arc pressure

Fig. 13 shows the arc pressure distribution characteristics under varying shielding gas compositions. During the hybrid welding process, the arc pressure is affected by the current density and plasma flow speed, as well as by the laser-induced metal plume. Therefore, the high-pressure areas of arc are located at the end of electrical tip and the laser-induced keyhole. When the shielding gas is Ar, the arc pressures peak at position A and C are 1004.3 Pa and 1012.8 Pa, respectively. As the laser rotates to position B and D, the arc pressure peak is concentrated only at the end of welding wire, and the arc pressure peaks are 772.8 Pa and 770.1 Pa, respectively. When the He content is 30%, the pressure peaks at positions A, B, C and D increased to 1093.2 Pa, 875.3 Pa, 1098.8 Pa and 885.3 Pa, respectively. When the shielding gas contains 50%He, the arc pressure at the middle position of the electrode and keyhole also increased. Meanwhile, the pressure peaks at each position further increased to 1132.1 Pa, 981.8 Pa, 1207.6 Pa and 985.2 Pa, respectively. It shows that as the proportion of He increases in the mixed shielding gas, the plasma density distribution becomes more concentrated in the direction of the welding wire, and the pressure value also increases. The overall size of hybrid plasma decreases with the increase of He content and laser-induced metal plume. At meanwhile, a larger plasma flow velocity and arc pressure are provided, which increases the penetration depth of rotating laser arc hybrid welding and effectively compensates for the limitations of this welding process method.

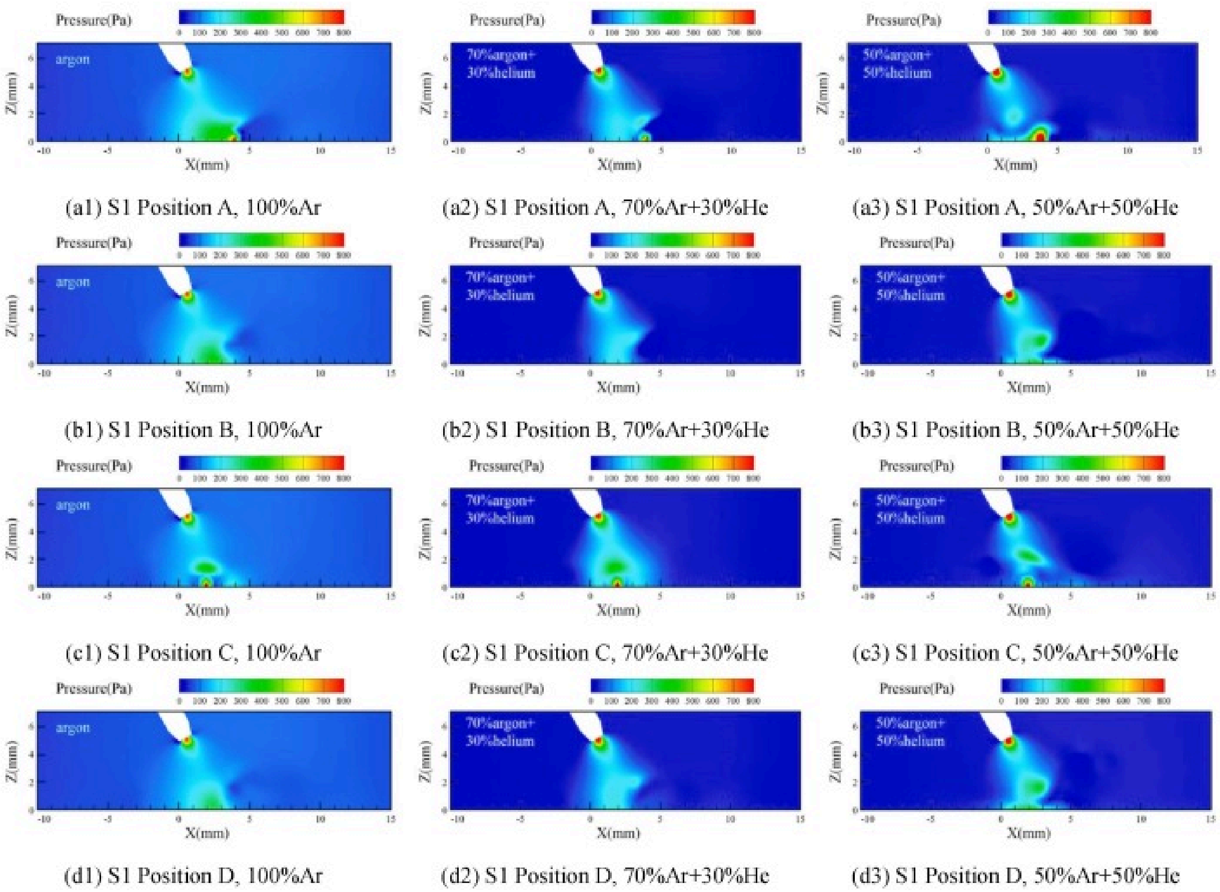


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Fig. 13. Longitudinal section plasma pressure at various positions.

## 4.4. Verification of model

Fig. 14 compares the calculated arc plasma geometries with the measurement data under different shielding gases. The arc length, height, and area obtained by calculations and experiments show consistent variations with the volume ratio of He, as shown in Fig. 15. The length and area of arc plasma decrease with increasing He content, and the decreasing trend is also roughly consistent. The arc height first increases and then decreases with increasing He content, with overall small changes in size due to the constant distance between the electrode and the surface of workpiece. It is revealed that the plasma shape obtained from simulation and experiment is basically consistent under different shielding gas composition, indicating the established model can accurately reflect the physical properties of plasma. In addition, it can be noticed that there are still some differences between the calculation cloud diagrams and the measured arc images. Due to the complexity of multi-physics field coupling in rotating hybrid welding, the model is simplified and lacks the high temperature properties of materials and shielding gases, which causes errors in the calculations to a certain extent.

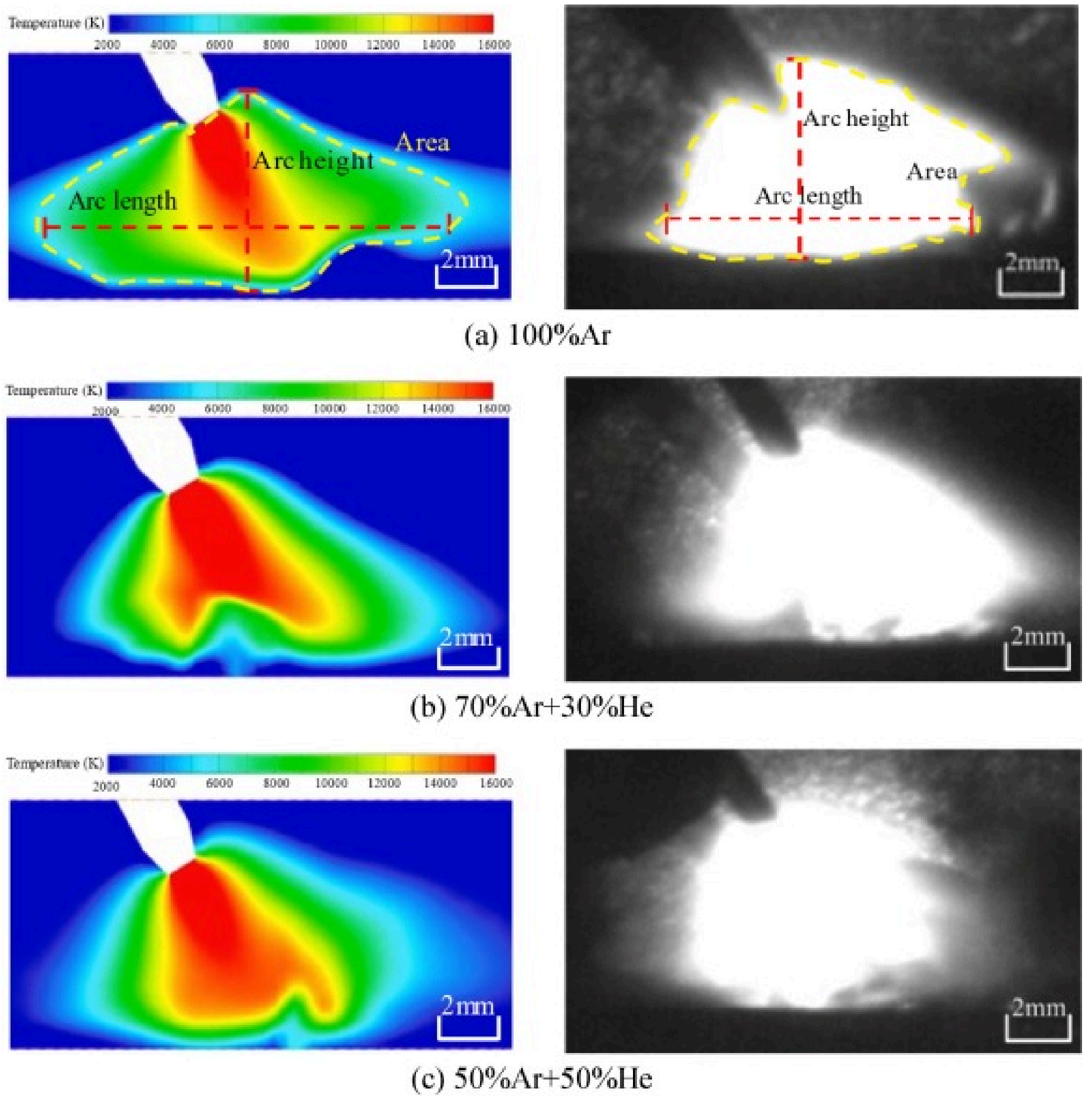


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Fig. 14. Schematic diagram of comparison between high-speed camera test and simulation results. (a) 100%Ar. (b) 70%Ar+30%He. (c) 50%Ar+50%He.

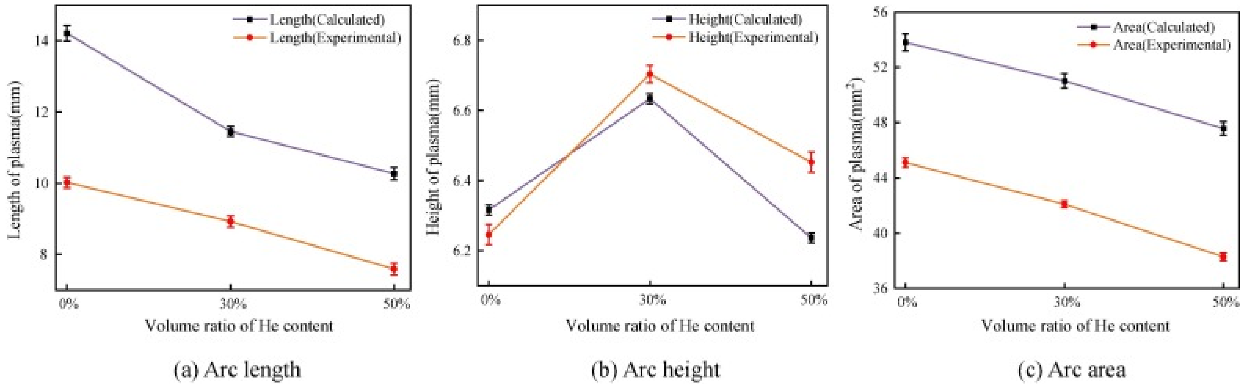


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Fig. 15. Comparison of calculated arc plasma dimensions with experimental data.

## 5. Conclusion

(1) The effects of shielding gas composition of 100%Ar, 70%Ar+30%He and 50%Ar+50% He on the physical properties of arc plasma in rotating laser+GMAW hybrid welding are investigated using a comprehensive model. It is evident that the shielding gas composition has a significant effect on the arc temperature, velocity, and pressure.

(2) As the proportion of He in the shielding gas increases, the overall morphology of arc obvious contraction and the arc plasma peak temperature decreases, the arc plasma high-temperature region is compressed, and the arc distribution is more concentrated, due to the higher ionization potential and thermal conductivity of He gas. The shielding effect of hybrid plasma is suppressed, and more laser energy reaches the surface of substrate, which helps to increase the weld penetration depth.

(3) With the He content increases, the arc plasma flow velocity increase, when the He content of 50%, the plasma flow state appeared obvious vortex and the flow velocity increases significantly, the flow state is chaotic. Although He gas can increase the penetration depth, its value is too high to stabilize the arc.

(4) The pressure peaks at different positions are located at the electrical tip and the keyhole. As the He ratio increases, the arc pressure values increase, and the arc pressure in the middle region of electrode tip and keyhole increases, which is conducive to concentrating the arc heat flow density.

## CRediT authorship contribution statement

**Yaowei Wang:** Writing – original draft. **Wen Liu:** Data curation. **Wenda Li:** Supervision. **Xiaoyang Su:** Software. **Wenyong Zhao:** Writing – review & editing. **Guoxiang Xu:** Project administration, Funding acquisition. **Jie Zhu:** Supervision.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgments

This work is financially supported by the National Natural Science Foundation of China (Grant No.52375340, 51975263).

## Data availability

Data will be made available on request.

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